

EE140 Final Design Project

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May 4 2026

1 Overview

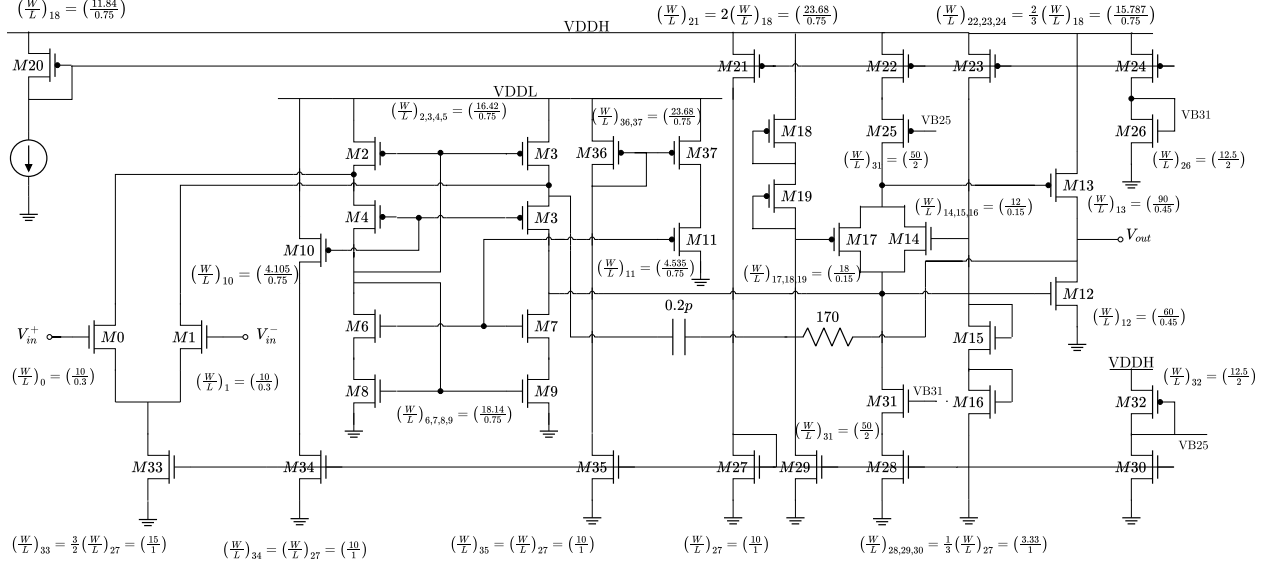


Figure 1: Full Op-Amp diagram. Note all devices are 2V devices and each PMOS body is tied to source and each NMOS body is tied to ground. All widths and lengths are given in micron. I_{bias} on left is set at $18.2\mu A$

The topology chosen for my LCD Driver operational amplifier is a two stage amplifier with a high gain folded cascode first stage and a high swing and high speed class AB second stage. Biasing is fully done with transistors and a single ideal current source. Power is drawn from a $VDDH = 1.8V$ and $VDDL = 1.1V$ rail. A $V_{cm} = 0.8$ on the input and $V_{ocm} = 0.9$ on the output were chosen as well.

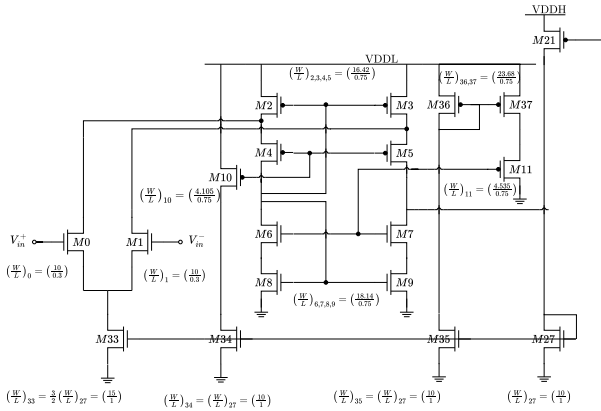


Figure 2: 1st Stage Folded Cascode

Folded Cascode Input Stage:

The folded cascode uses NMOS2V inputs and is powered and biased fully off the $VDDL = 1.1V$ rail. Both the NMOS and PMOS cascode are biased using the wide swing cascode biasing topology, with 2 additional needed biasing legs.

Class AB Output Stage:

The Class AB Output Stage is implemented with an inverter based design and Monticelli biasing for high swing and good process and temperature tracking. This stage is powered purely off of $VDDH = 1.8V$ rail. The Monticelli topology is pulled straight from the Hogervorst paper with the ideal current sources implemented using cascoded current mirrors. Each cascoded current mirror is implemented using a wide swing cascode to save headroom on output transistors.

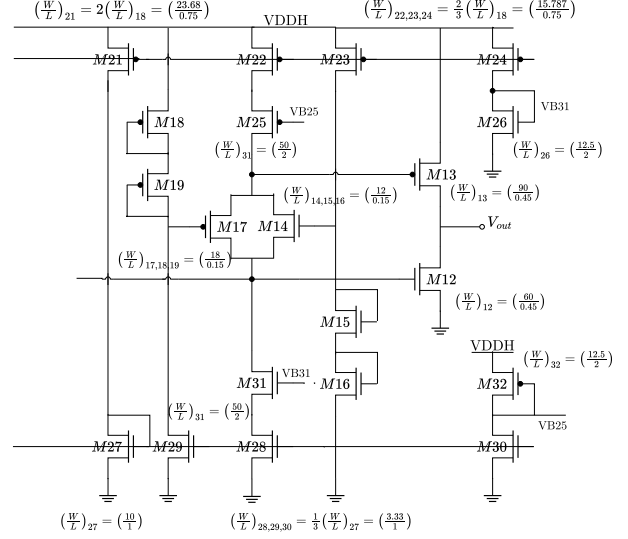


Figure 3: 2nd Stage Class AB with Monticelli Biasing

Compensation:

For better control of phase margin and faster settling Ahuja compensation is used by tying a compensator capacitor between the amplifier output and the PMOS cascode of the first stage. A compensation resistor is additionally used in series.

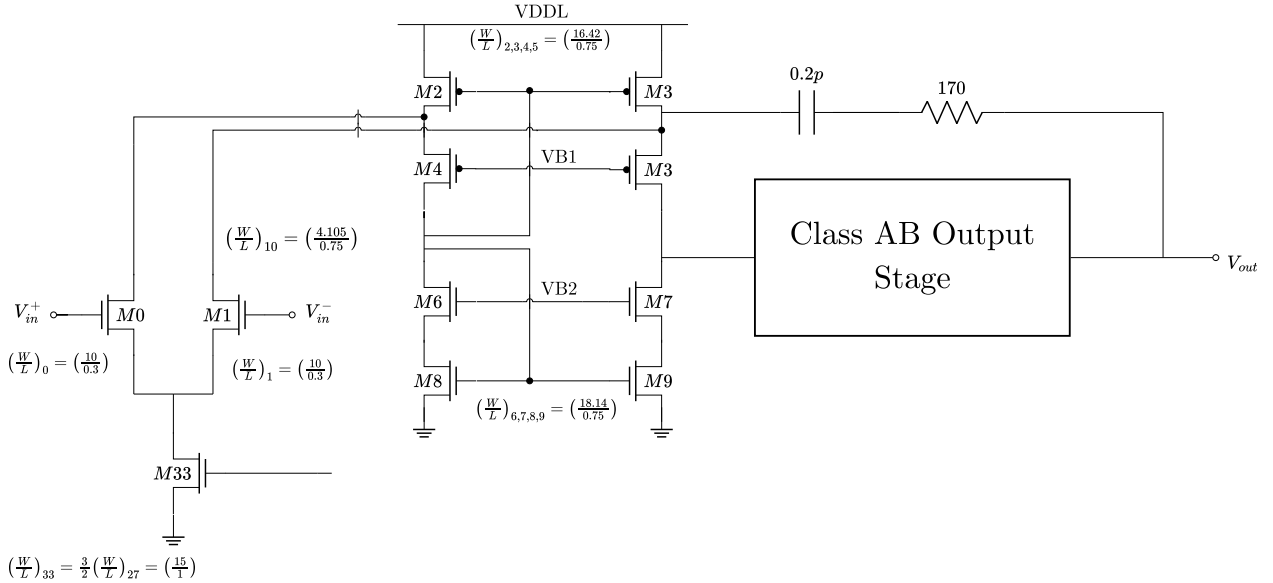


Figure 4: Ahuja Compensation

Table 1: Comparison of Simulated Design Performance versus Specifications.

Parameter	Simulated Design	Design Specs (240A)
Technology	gpdk045	gpdk045
Power Supplies (VDDL)	$VDDL = 1.1$	$VDDL \leq 1.1$
Power Supplies (VDDH)	$VDDH = 1.8$	$VDDH \leq 1.8$
Closed Loop DC Gain	2	2
Load	$150\Omega, 60pF$	$150\Omega, 60pF$
Settling Time	$119ns$	$\leq 180ns$
Total Error	$\leq 0.2\%$	$\leq 0.2\%$
Output Voltage Swing	$\geq 1.4V$	$\geq 1.4V$
Max Current Mirror Ratio	4.5	10
Max Total Capacitance	$0.2pF$	$10pF$
Max Total Resistance	170Ω	$100k\Omega$
CMRR at DC	$77.65dB$	$\geq 65dB$
PSRR at DC (VDDL)	$57.62dB$	$\geq 50dB$
PSRR at DC (VDDH)	$53.45dB$	$\geq 50dB$
Phase Margin	65.07°	$\geq 45^\circ$
Power Consumption	$1.052mW$	$\leq 1.5mW$

As displayed, the amplifier meets design specifications in every category with a power consumption of $1.052mW$.

2 Design

2.1 Hand-Calculations

Remark: the major specifications to hit are settling time, PSRR, and CMRR. Hitting settling is handled by pole placement and gain, while PSRR and CMRR is handled by topology.

Starting with settling, I partitioned my static and dynamic error as 50/50. Doing this we can find the minimum open loop gain for our minimum static error:

$$\epsilon_s = \frac{1}{1 + RR} = \frac{1}{1 + \beta A_{OL,0,min}} \rightarrow A_{OL,0,min} = \frac{1}{\beta} \left(\frac{1}{\epsilon_s} - 1 \right) = \frac{1}{\frac{1}{3}} \left(\frac{1}{0.001} - 1 \right) = 2997$$

Now consider our dynamic linear settling error. Using the Laplace transform where we insert in $V_{in}(s) = \frac{\Delta}{s}$ as our step input, we can use the dominant pole approximation of $\tau = \frac{1}{2\pi f_{3dB}}$ to find:

$$\epsilon_d = e^{-\frac{t_s}{\tau}} = e^{-t_s 2\pi f_{3dB}} \rightarrow f_{3dB} = \frac{-\ln(\epsilon_d)}{2\pi t_s} = \frac{\ln(\frac{1}{\epsilon_d})}{2\pi t_s}$$

From our spec we know we need to drive $272 * 340 * 60$ pixels per second. With the goal to settle 1/4 within linear and 3/4 in slew we can calculate:

$$f_{3dB,closed} = \frac{\ln(\frac{1}{\epsilon_d})}{2\pi t_s} = \frac{\ln(\frac{1}{0.0001})}{2\pi(180ns * \frac{1}{4})} = 32.6MHz \rightarrow f_{p1} = \frac{f_{3dB,closed}}{A_{0,OL}\beta} = 32.6kHz$$

Using this we can find a very important characteristic, our loop crossover frequency:

$$f_c = A_{0,OL}\beta f_{p1} = f_{3dB,closed} = 32.6MHz$$

I also choose to aim for a phase margin of 60 degrees, creating underdamped settling. Assuming an early dominant pole, I can solve for the equation:

$$\phi_m = 90^\circ - \arctan\left(\frac{f_u}{f_{p2}}\right) \rightarrow f_{p2} = \frac{f_u}{\tan(90^\circ - \phi_m)} = \frac{f_u}{\tan(30^\circ)} \approx 56.4MHz$$

Additionally, by using block ideal G_m and R_o values with Ahuja compensation we can find two poles and one zero given with the equations:

$$\omega_{p1} = -\frac{1}{G_{m2}R_{o1}R_{o2}C_c} \quad \omega_{p2} = -\frac{G_{m1}R_{o2}G_{m,cascode}}{C_L} \quad \omega_{z1} = -\frac{1}{C_c(\frac{1}{G_{m,casc}} + R_z)}$$

Starting with $C_c = 1pF$ (which will be changed later) and $C_L = 60pF$, and knowing $G_{m,cascode} = G_{m1}$ for a folded cascode it can be shown that we can find important design values for each stage as such:

$$G_{m1} = A_{0,OL}C_c\omega_{p1} = 614\mu S \quad G_{m2} = \frac{\omega_{p2}C_L}{G_{m1}R_{o2}} = 3.5mS \quad R_{o1}R_{o2} = \frac{A_{OL,0}}{G_{m1}G_{m2}} = 1.4G\Omega$$

Note that these values were calculated with the assumption the folded cascode would have ≈ 100 gain and the class AB output stage would have the rest. Using these values and assumptions we find:

$$R_{o1} \approx 140k\Omega \quad R_{o2} \approx 10k\Omega$$

To deal with slew rate, we can approximate that with a small compensation capacitor, we only care about output stage slew. We can calculate the min current for our given slew as follows (assuming we partition 75% of dynamic settling to slew):

$$I_{out} = \frac{V_{step}}{t_{slew}} C_L = \frac{1.4}{135n} 60p = 0.6mA$$

Remark: this I_{out} is used for rising and falling slew as the capacitor seen is the same.

2.2 Topology Choice

For the first stage, we can see from the hand-calculations I need a high gain stage with high output impedance. I also need high CMRR and PSRR. The two best topologies researched were the telescopic cascode and the folded cascode (which I decided ended up choosing to design around). The tradeoffs for the folded are below:

- **Pros:** High ICMR, good headroom, lower $V_{out,Q}$
- **Cons:** slightly lower gain, higher complexity

Due to the use of Monticelli biasing (discussed later), the folded cascode was much better as a lower quiescent voltage and a more resistance topology to leakage was preferred. The higher headroom also allowed me to use $VDDL = 1.1V$ to reduce power.

For the folded cascode I designed around the following small signal equations:

$$G_m = g_{m0,1} \quad R_o = (g_{m4,5} r_{o4,5} (r_{o2,3} \parallel r_{o0,1})) \parallel (g_{m6,7} r_{o6,7} r_{o8,9})$$

For the second stage, we need reasonable gain, high swing, and high current draws on rising/falling edges. In turn, I chose an inverter based class AB amplifier output stage. Class AB allows for a current push-pull, dealing with slew rate. Due to specs requiring a 1.4V swing, I chose to use a $VDDH = 1.8V$ rail. Here are the following small signal equations:

$$G_m = g_{m12} + g_{m13} \quad R_o = r_{o12} \parallel r_{o13}$$

2.3 Biasing

The first stage is biased fully with wide swing cascodes. The middle V_{bias} rails are sized using $\frac{1}{4} \left(\frac{W}{L}\right)_N$ and $\frac{1}{4} \left(\frac{W}{L}\right)_P$ biasing to achieve the $V_{th} + 2\Delta V$ on the cascode transistors.

The input tail currents were sized for $I_{tail} \approx I_{top}$ for $I_{bot} \approx \frac{1}{2} I_{top}$ to bias lower transistors. For simplicity I additionally sized all NMOS cascode transistors the same and all PMOS cascode transistors the same.

The second stage is biased using Monticelli biasing. The sizings for the level shifters were used with a ratio of $\frac{1}{5} \left(\frac{W}{L}\right)_N$ and $\frac{1}{5} \left(\frac{W}{L}\right)_P$ with regards to the output NMOS and PMOS. The $2\Delta V + 2V_{th}$ biasing for those level shifters were designed using two diode connected transistors with identical sizing and currents.

The ideal current sources were emulated using regular current mirrors on the biasing legs and a cascoded current mirror on the main level shifting leg. There was a tradeoff between headroom and PSRR, but to increase my PSRR I decided the cascode was worth it. Each cascode is also biased using $\frac{1}{4} \left(\frac{W}{L}\right)$ wide-swing biasing techniques as mentioned earlier.

2.4 Scripts

To get actual sizing values and current, I used gm/Id design methodology scripts for main amplifier transistors and current mirrors.

For the first stage I fixed an $L = 0.3\mu m$ for better g_m pole placement on my input and $L = 0.75\mu m$ for the cascode transistors themselves for higher gain. The goal of the first stage was high gain, so weaker inversion gm/Id's were swept over to size all transistors. The script searched and filtered for gm/Id's with results matching or exceeding hand-calculation results.

Additional manual sorting was done to find a combination of a low I_{bias} for lower power and a $V_{out,Q}$ high enough to keep the output NMOS on, but low enough to reduce nominal I_d draw on the output stage.

For the second stage I initially fixed an $L = 0.15\mu m$ for high speed (which will change later). I also fixed the $V_{gs,n}$ as the $V_{out,Q}$ of the first stage for all sweeps. I swept over stronger inversion than my first stage for a greater mix of speed and gain on my output transistors. The script searched and filtered for gm/Id's matching or exceeding hand-calculation results, with biasing transistors chosen as per (2.3 Biasing).

For the current mirrors I designed for higher L values for higher r_o and therefore higher CMRR and PSRR. The gm/Id was swept for stronger inversion for better r_o and current matching and was filtered for closer V_{ds} and high r_o values.

2.5 Simulation and Iteration

Simulation always starts on open loop AC test-benches to find DC operating points. Due to the powerful topology chosen, the priority in simulating DC operating points was to ensure all transistors were saturation, as most sizings the script outputted had more than enough gain and settling speed. DC operating points also allowed me to match Cadence values to hand-calculated values.

After getting AC specifications and goals correct—such as gain, phase margin, CMRR, and PSRR—I moved onto transient analysis to match settling requirements. The main issue ran into here is leakage into my first stage caused by DC bias point imbalances on the V_{ds} of my first stages two legs caused by the Monticelli setting a different DC bias point.

Additionally on a more micro level, specific iterations and changes will be discussed deeper (4 Discussion), but the main iterations I worked through were scaling current higher, increasing lengths, and deciding pole placement using the compensation capacitor.

1. **Current:** I_d/W was held constant, but I_d and W were scaled accordingly to increase G_{m1} for pole placement and to handle a higher slew rate. Current was also increased from a $I_d/5$ ratio to a $I_d/3$ ratio for the Monticelli biasing branches to establish a more stable DC operating point and improve PSRR through increasing transconductance.
2. **Lengths:** I struggled with PSRR, so tail area was increased. Cascodes on Monticelli current sources were also implemented for similar reasons.
3. **Pole Placement:** To decrease phase margin to 65° and decrease potential input stage slew, I decreased C_c to $0.2pF$.

Remark: most changes elsewhere were done purely in Cadence as changes were minimal.

The final big iterations made were topology changes. I did similar design processes for these topologies over the past 5 weeks in this order below:

1. I started with a telescopic input into source follower class AB output with miller compensation for simplicity within spec.
2. I went to telescopic input into source follower class AB output with Ahuja compensation for better pole placement and phase margin
3. I then moved to telescopic input into Monticelli class AB output with Ahuja compensation for a more practical real world design for process tracking.
4. I ended with the folded input into Monticelli class AB output with Ahuja compensation, as the Monticelli leaked too much current into my telescopic creating too wide of an imbalance between differential legs.

Scripts were tweaked and re-run each iteration and extra biasing was added appropriately.

3 Transistor and Bias Summary

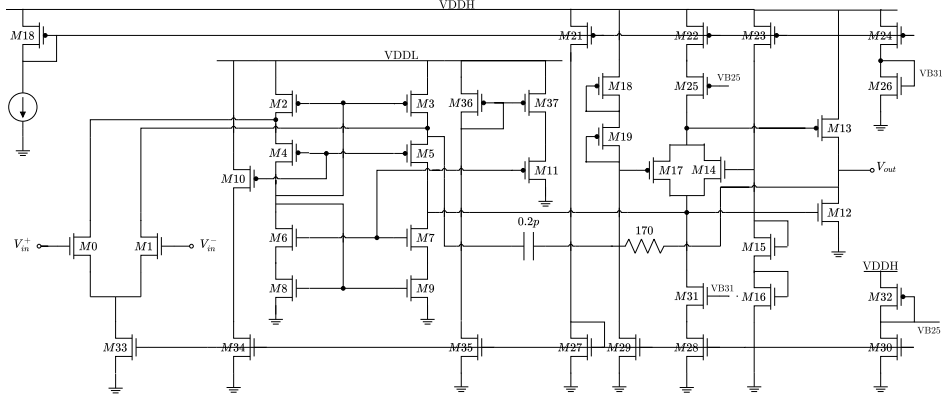


Figure 5: Full schematic without sizings for convenience

	W μm	L μm	$ I_d $ μA	$ V_{gs} $ mV	g_m μS	g_{ds} μS
M0	10	0.3	28.0	578	443	13.4
M1	10	0.3	28.0	578	443	13.4
M2	16.42	0.75	50.5	584	599	20.8
M3	16.42	0.75	50.4	584	598	21.0
M4	16.42	0.75	22.5	528	385	9.37
M5	16.42	0.75	22.4	530	380	13.1
M6	18.14	0.75	22.5	543	364	15.2
M7	18.14	0.75	22.5	541	365	12.9
M8	18.14	0.75	22.5	516	361	19.6
M9	18.14	0.75	22.5	516	362	19.5
M10	4.105	0.75	83.4	610	856	22.9
M11	4.535	0.75	78.1	764	464	11.7
M12	60	0.45	339	580	4159	80.8
M13	90	0.45	136	519	2518	41.9
M14	12	0.15	1.5	463	38.8	1.00
M15	12	0.15	14.1	555	277	10.7
M16	12	0.15	14.1	488	275	13.2
M17	18	0.15	6.85	445	178	4.52
M18	18	0.15	14.7	482	350	10.8

	W μm	L μm	$ I_d $ μA	$ V_{gs} $ mV	g_m μS	g_{ds} μS
M19	18	0.15	14.7	482	350	10.8
M20	11.84	0.75	18.2	530	303	5.33
M21	23.68	0.75	40.8	530	648	9.14
M22	15.79	0.75	8.35	530	120	63.8
M23	15.79	0.75	14.1	530	230	3.71
M24	15.79	0.75	16.0	530	251	3.51
M25	50	2	8.34	442	200	3.01
M26	12.5	2	16.0	514	241	5.85
M27	10	1	40.8	610	419	9.20
M28	3.33	1	8.25	610	67.4	66.8
M29	3.33	1	14.7	610	147	2.40
M30	3.33	1	15.6	610	154	1.89
M31	50	2	8.43	430	164	3.22
M32	12.5	2	15.63	518	279	4.86
M33	15	1	56.0	610	584	40.0
M34	10	1	76.6	610	801	39.7
M35	10	1	83.4	610	856	22.9
M36	23.68	0.75	83.4	593	928	18.2
M37	23.68	0.75	78.1	593	878	26.5

Table 2: Transistor and DC Operating Points Summary

4 Discussion

4.1 AC Performance

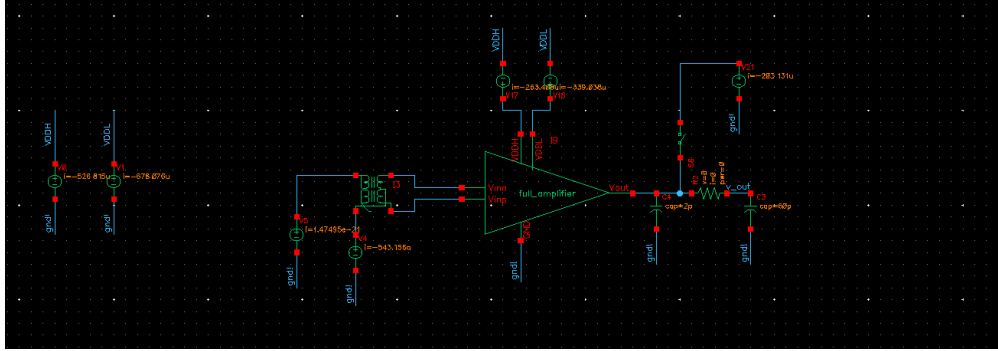


Figure 6: AC testbench for calculating CMRR and PSRR

The testbench here has input AC on the VDDH and VDDL rails as well as on the common mode rails. This allows for calculation of the CMRR and PSRR.

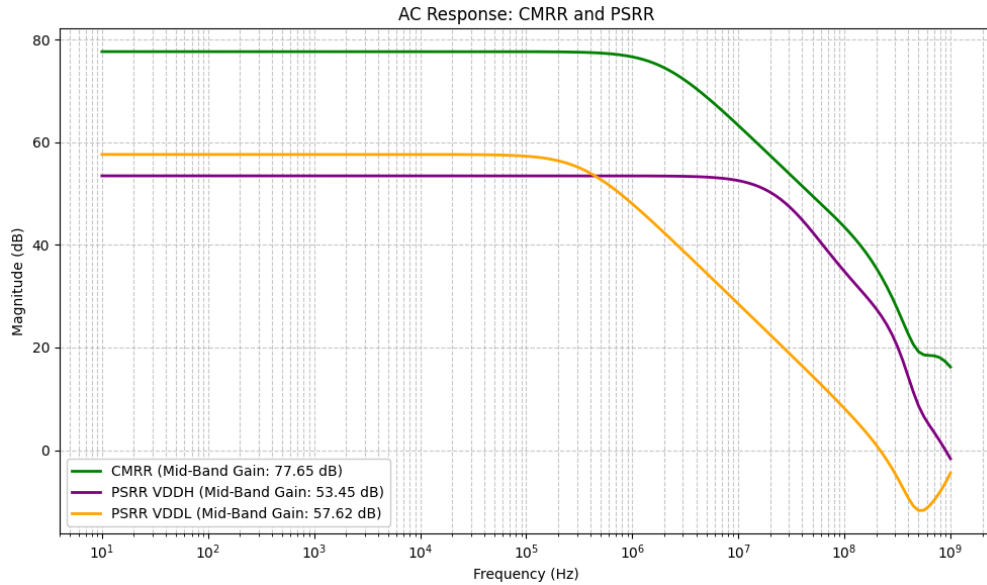


Figure 7: CMRR and PSRR versus Frequency Graph

The CMRR and PSRR are within specifications as seen in the bode plot. CMRR was easy to hit with the given topology due to source degeneration from the folded cascode and the design choice to increase tail transistor lengths.

The harder specification to hit was PSRR. The main interesting optimizations that had to be made was trying to increase output resistance between the input nodes and main signal nodes for the differential signal. This was done by a lot of iterating with larger lengths as well as adding the cascodes on the Monticelli current mirrors.

The theory behind the cascodes was for two major reasons:

1. I wanted to reduce error from static gain. I noticed when simulating that adding the Monticelli reduced my DM gain from $\approx 80dB$ to around $\leq 60dB$. I realized this was because the output impedance seen from my 1st stage was reduced from some $g_m r_o^2$ term from my telescopic (at the time) to a r_o term from the Monticelli current mirror. Adding a cascode to the current mirror increasing the seen impedance to a $g_m r_o^2$.
2. I wanted to increase PSRR for VDDH as it was below specifications at $\approx 40dB$. This was first achieved from the previous point as increasing DM gain by definition increases PSRR. Additionally, the cascode shields the VDDH rail from the second stage input through VDDH seeing an additional transistor in the way.

Additionally, I want to note that with an earlier iterations before using the folded cascode and Monticelli biasing I was able to hit the original PSRR spec of $\geq 65dB$. I imagine the PSRR was very good on the input as I used PMOS input and we only cared about PSRR+ not PSRR-. Similarly the source follower on the output stage is also very resistant. The Monticelli addition ended up killing this PSRR, so I settled in my final design to have a PSRR below $65dB$.

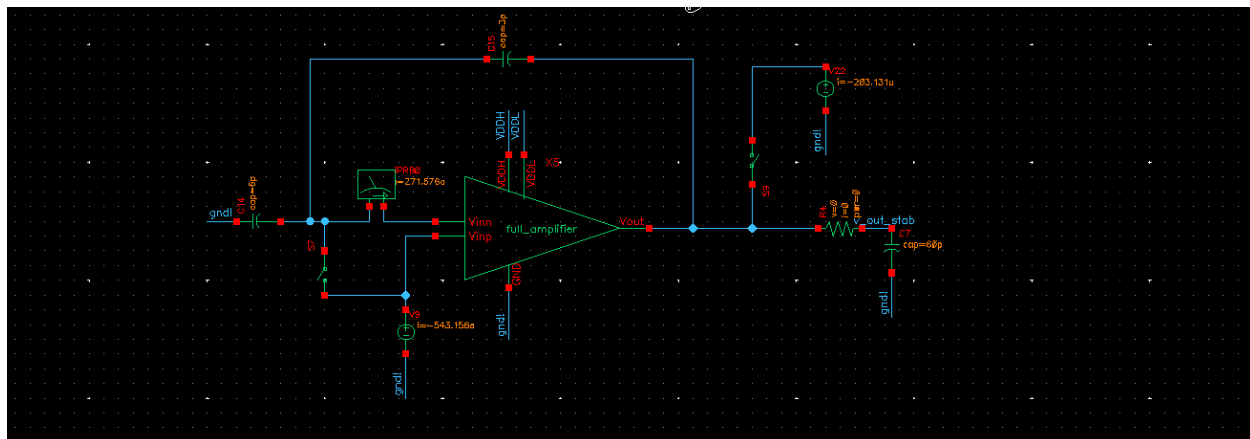


Figure 8: AC testbench for loop gain, stability, and phase margin

The testbench here has a switch allowing for calculations of phase and loop gain. I used this to verify pole placement and stability.

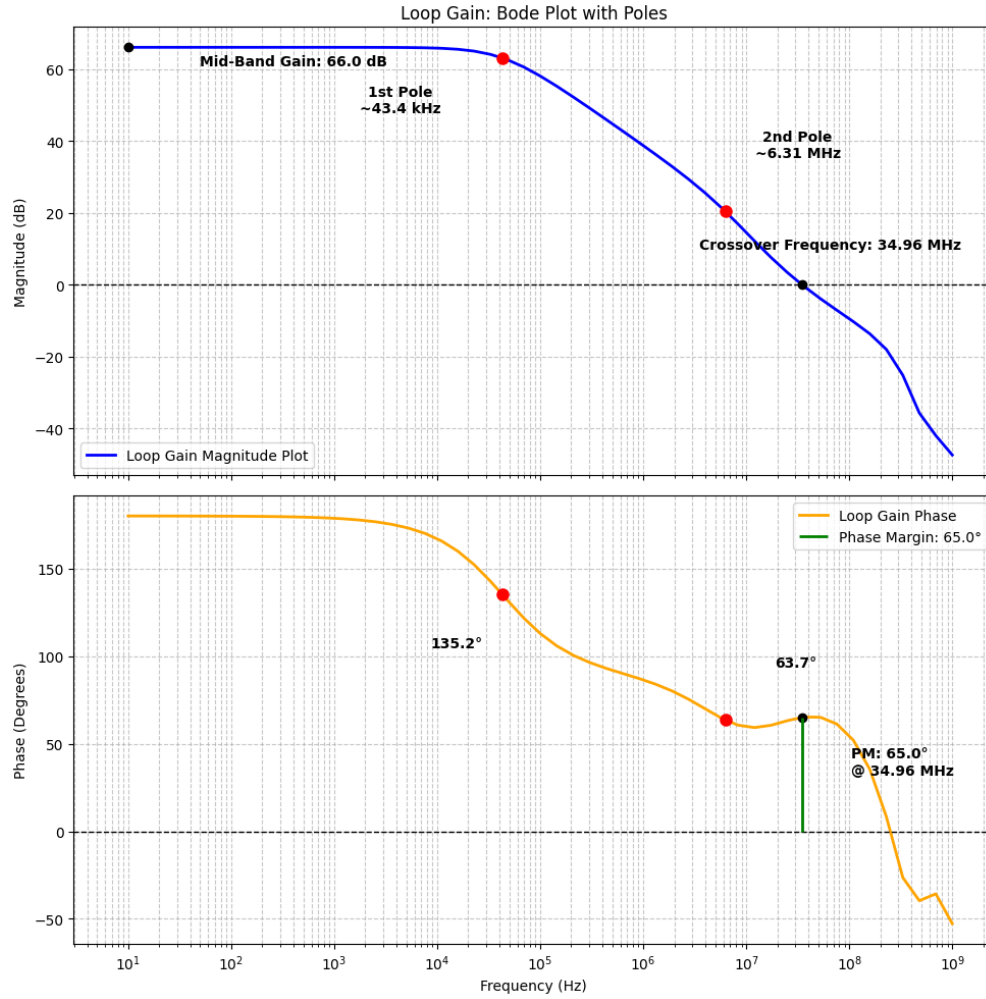


Figure 9: AC testbench for loop gain, stability, and phase margin

The plot shows the pole placement as well as the phase margin of my circuit. As mentioned before, the poles were achieved purely based on initial scripting and compensation capacitor pole splitting.

As mentioned before, the phase margin spec was ignored in favor of trying to place phase margin in an ideal spot for under-damped settling. Hence as hand calculated, we see a phase margin in the range of 60°

Table 3: Comparison of Simulated Design Performance versus Hand Calculations.

Parameter	Simulated Design	Hand-Calculations
G_{m1}	$443\mu S$	$614\mu S$
G_{m2}	$6.68mS$	$3.5mS$
R_{o1}	$\approx 250k\Omega$	$140k\Omega$
R_{o2}	$8.2k\Omega$	$10k\Omega$
f_{p1}	$43.4kHz$	$32.6kHz$
f_{p2}	$6.31MHz$	$56.4MHz^*$
A_v	$75dB$	$\geq 70dB$

*Note: The simulated f_{p2} is heavily shifted due to the LHP zero from Ahuja compensation. See Section 4.1 for full analysis.

Viewing Cadence DC operating points and phase margin simulations we get a comparison of hand calculations to actual results.

We can see very close similarities in most parameters, sanity checking the reliability of the scripts. However, I want to highlight the main differences in second pole placement caused by Ahuja compensation.

The Ahuja compensation changes the second pole to be closer by a factor of $G_{m1}R_{o2}$. However, the closer pole is offset by a LHP zero created by the capacitor. In turn, our pole is earlier, but our phase is supplemented by the zero keeping the phase higher. This is why in the plot it looks as if the phase is constant right after the second pole.

I will also note that the nulling resistor (which was a remnant from a miller compensation topology) ended up not being helpful nor hurtful to my design, and in retrospect could probably be removed. The reasoning why comes from this equation from earlier:

$$\omega_{z1} = -\frac{1}{C_c(\frac{1}{G_{m,casc}} + R_z)}$$

As you can see the nulling resistor is in series with the G_m of the cascode. In turn, the zero is already nulled and pushed into the LHP.

In total, we have easily hit our static error design decision with a higher gain, and we have placed our poles generally where we want them for correct phase margin.

4.2 Transient Performance

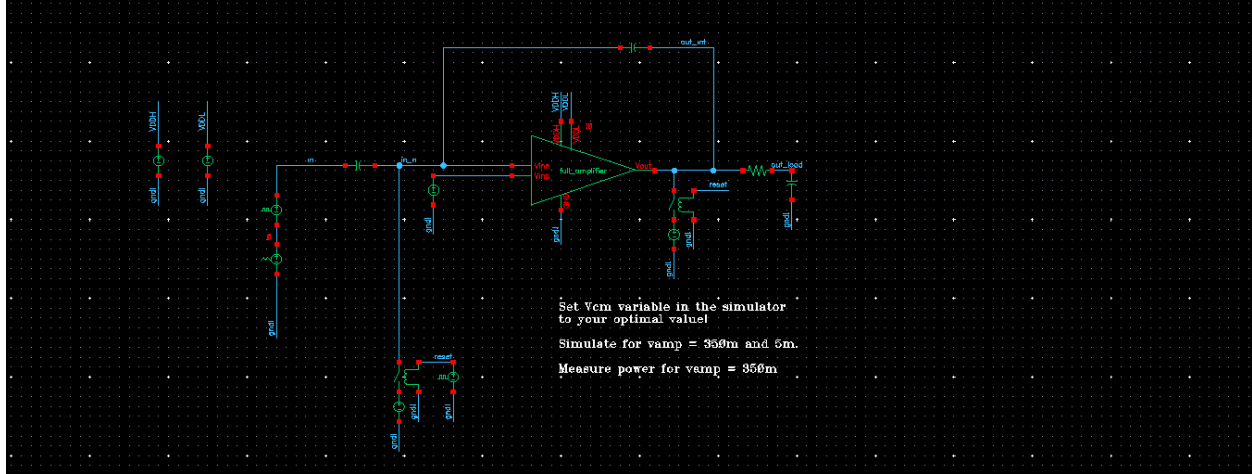


Figure 10: Transient testbench for settling time, current draw, and transient analysis

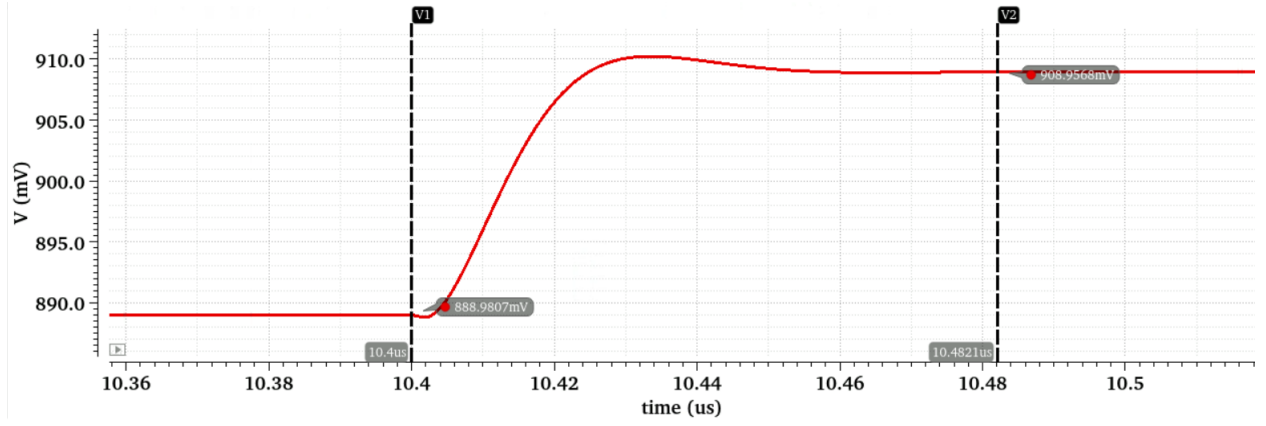
The testbench here has a switch allowing a setting of common mode output as well as the full closed loop gain of 2 from the spec. It is used for all transient analysis.

Test	Output	Spec	Weight	Pass/Fail	Min	Max	5mV	350mV
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
project_lib_tb_tr...	vout							
project_lib_tb_tr...	vout_load							
project_lib_tb_tr...	avg_pow_vddi				372.9u	372.9u	372.9u	372.9u
project_lib_tb_tr...	avg_pow_vddh				589.8u	679.1u	589.8u	679.1u
project_lib_tb_tr...	avg_pow_tot	< 2.75m		pass	962.7u	1.052m	962.7u	1.052m
project_lib_tb_tr...	err_high							
project_lib_tb_tr...	rise_time				82.08n	119.2n	82.08n	119.2n
project_lib_tb_tr...	err_low							
project_lib_tb_tr...	fall_time				66.72n	81.78n	81.78n	66.72n
project_lib_tb_tr...	settle_time	< 180n		pass	82.08n	119.2n	82.08n	119.2n
project_lib_tb_tr...	/R0/PLUS							
project_lib_tb_tr...	/C5/MINUS							
project_lib_tb_tr...	/W1/N+							
project_lib_tb_tr...	/I0/Vinp							
project_lib_tb_tr...	/I0/Vinn							
project_lib_tb_tr...	/I0/GND							
project_lib_tb_tr...	/I0/VDDL							
project_lib_tb_tr...	/I0/VDDH							

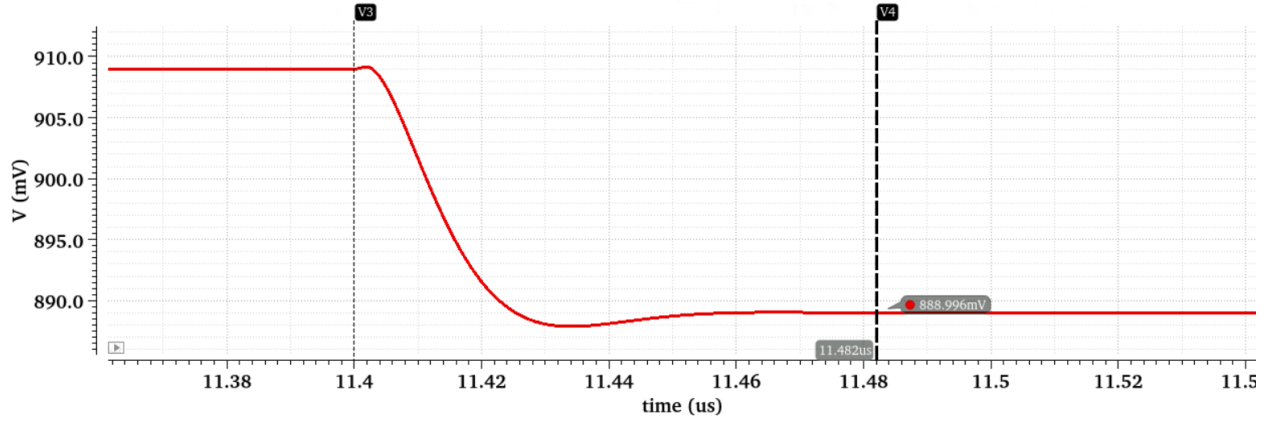
Figure 11: Transient Maestro screenshot for settling time and power specifications

As shown, the power and settling times as seen in Maestro hit the specifications. As I walk through each individual settling plot I will discuss specific challenges.

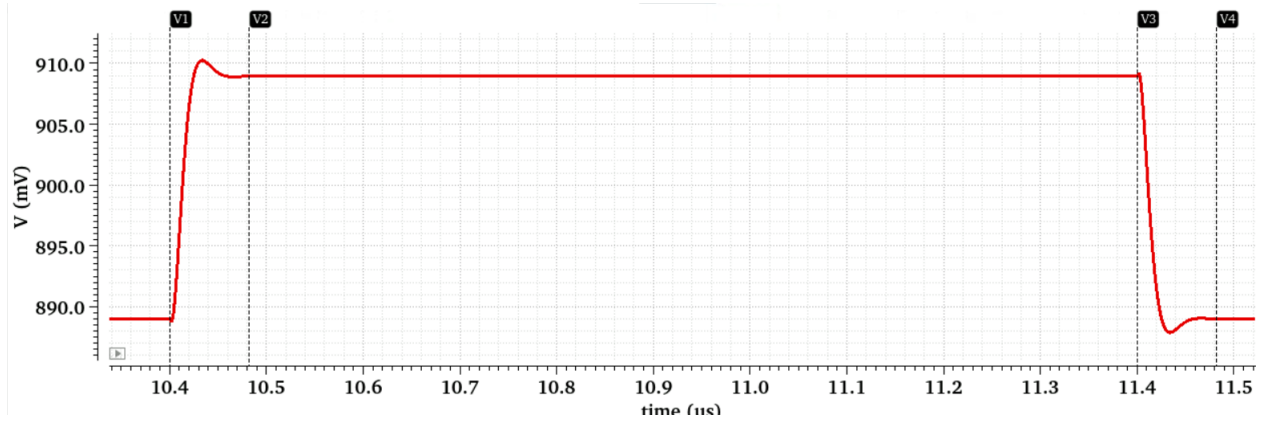
However generally, an improvement that could be made in this category is power draw. I got this topology within specifications very late and in turn didn't have time to optimize it.



(a) Rising Waveform at $V_{out,load}$ 5mV

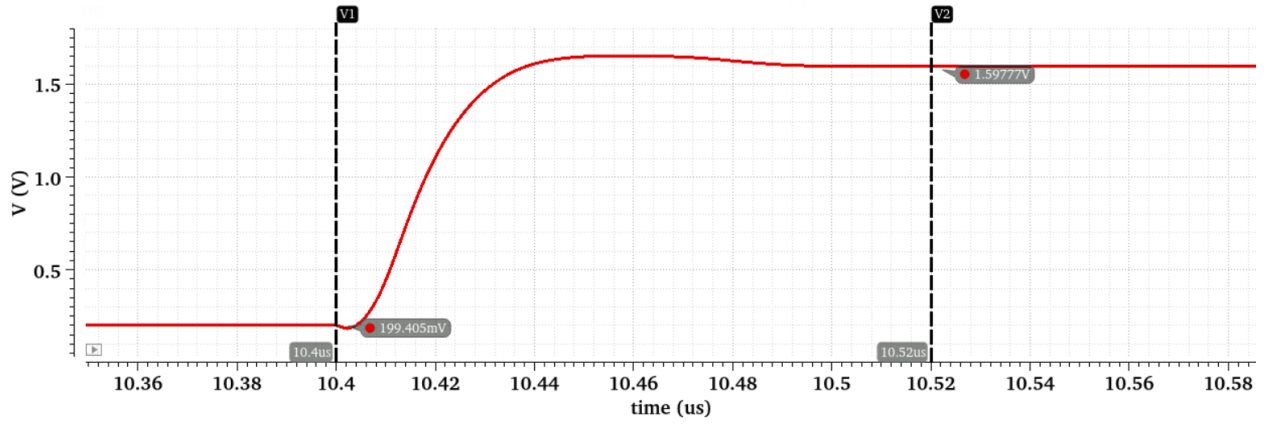


(b) Falling Waveform at $V_{out,load}$ 5mV

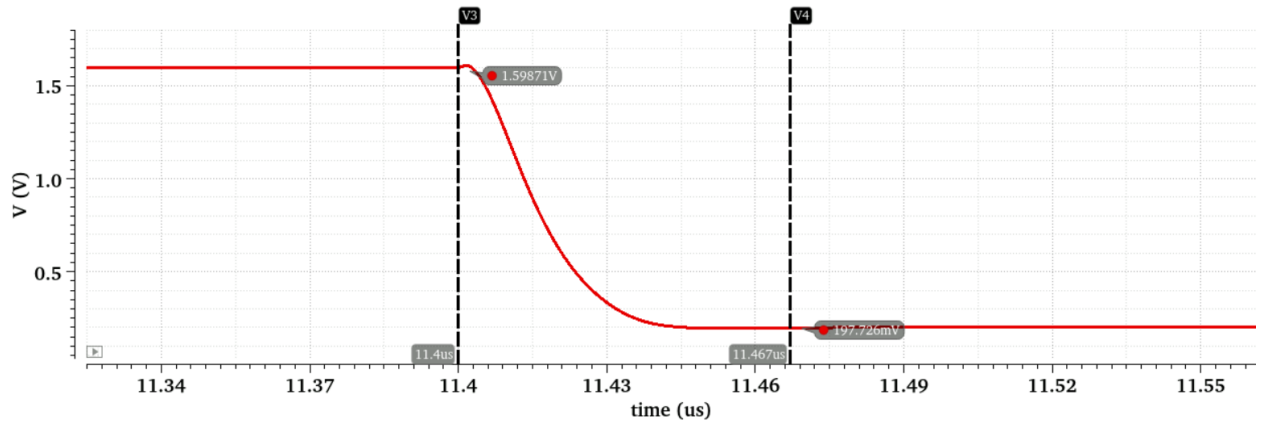


(c) Full Cycle Waveform at $V_{out,load}$ 5mV

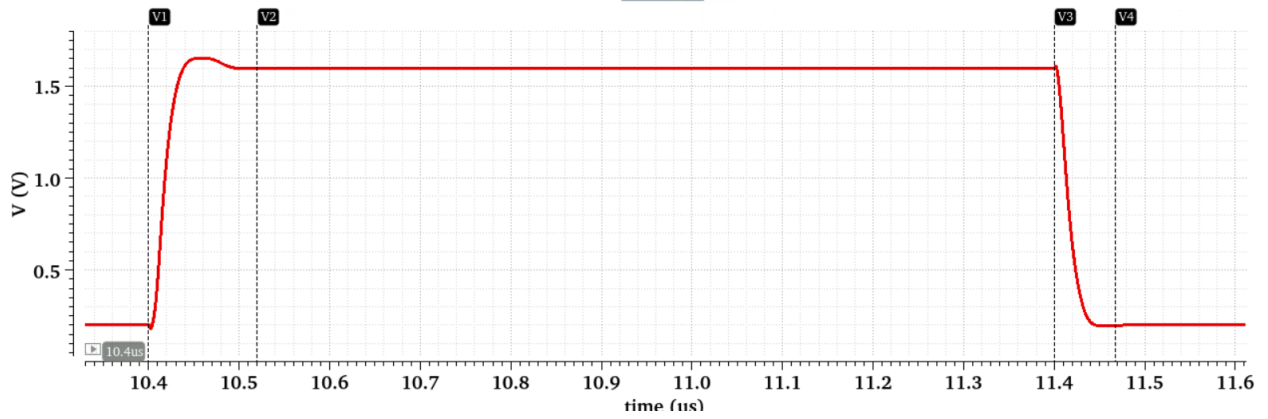
Figure 12: Transient settling response of the amplifier for a 5mV step.



(a) Rising Waveform at $V_{out,load}$ 350mV



(b) Falling Waveform at $V_{out,load}$ 350mV



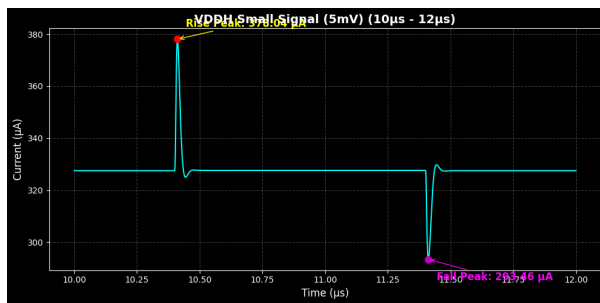
(c) Full Cycle Waveform at $V_{out,load}$ 350mV

Figure 13: Transient settling response of the amplifier for a 5mV step.

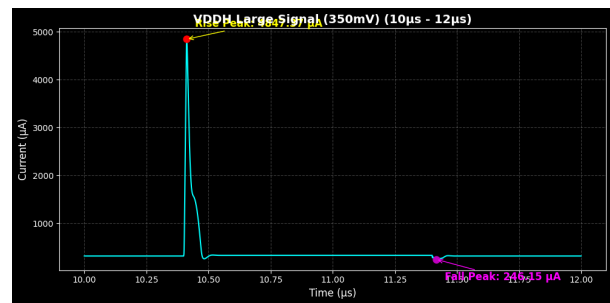
Above on the previous two pages are the rising and falling for $V_{out,load}$ at 5mV and 350mV. The rising and falling looks like it is a good mix of slewing and linear settling. This is as expected due to a good phase margin and a reasonable current draw.

The main discussion on these waveforms is the main problem I dealt with in the project as a whole which is leakage. Due to Monticelli biasing there was a large V_{ds} difference in my telescopic cascode back when that was my topology. The main reason is my telescopic was trying to set my middle telescopic node at 400mV and my monticelli was trying to pull it up to 600mV. This caused a large branch mismatch leading to leakage.

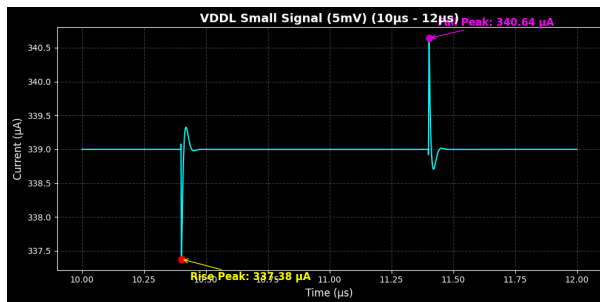
The solve as mentioned before was switching topologies to a folded cascode that set the quiescent current closer to what the Monticelli biasing wanted. More discussion on this mentioned earlier.



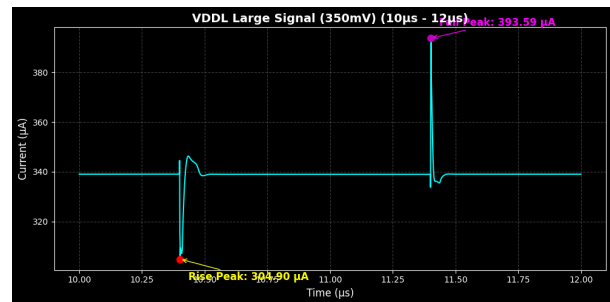
(a) VDDH 5mV Current



(b) VDDH 350mV Current



(c) VDDL 5mV Current

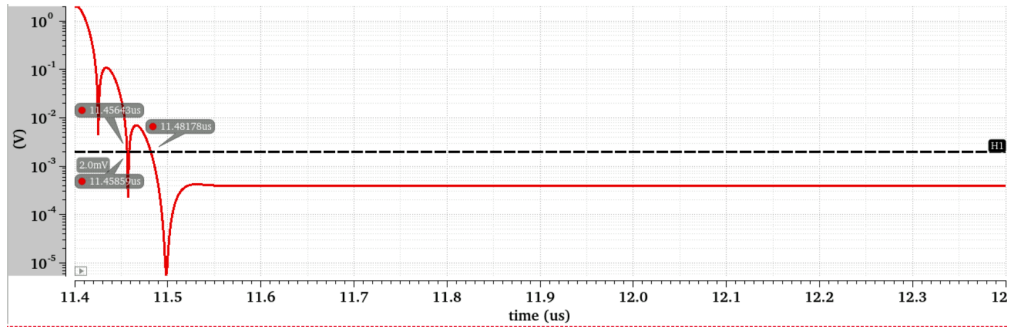


(d) VDDL 350mV Current

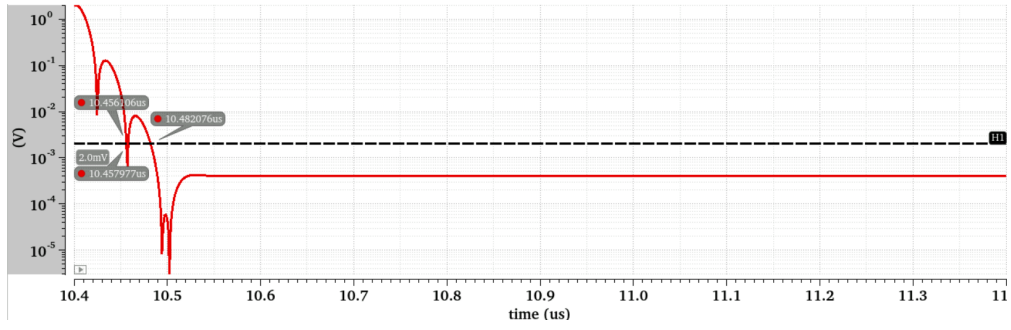
Figure 14: Current Draw For Different Inputs

The most interesting image to discuss is the "VDDH 350mV Current" and how it deals with peaks and valleys. The higher current draw is due to the class AB demanding a higher current to deal with output slew.

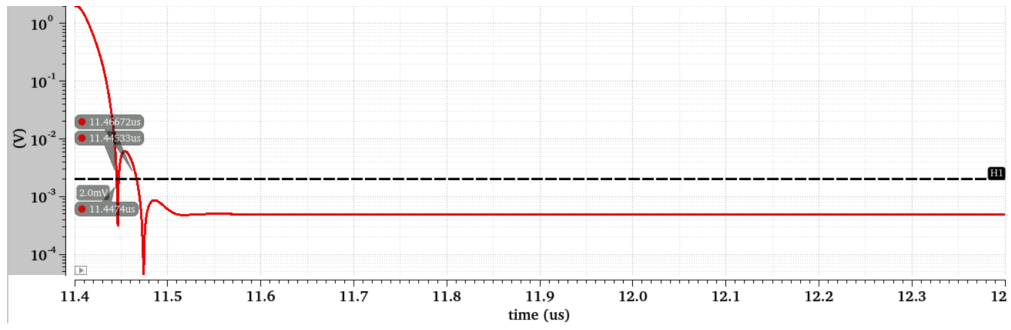
The positive current spike is considerably higher compared to the negative current spike due to an imbalance in the class AB output stage due to different sizings in the PMOS and NMOS. This doesn't effect the amplifier hitting specification, but it is a note for future design.



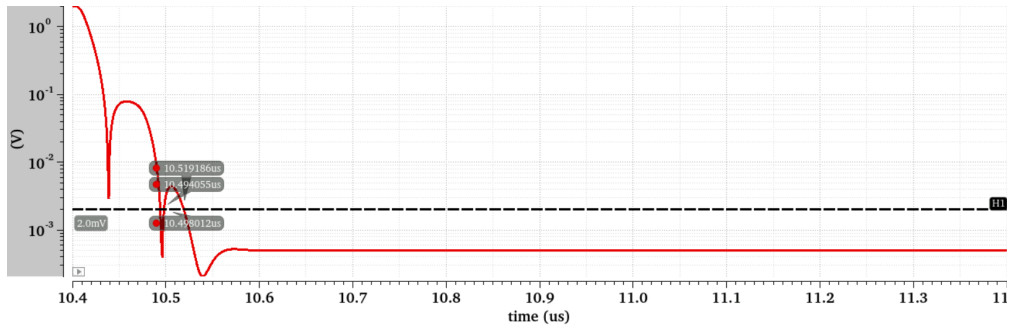
(a) 5mV Falling Error



(b) 5mV Rising Error



(c) 350mV Falling Error



(d) 350mV Rising Error

Figure 15: Rising and Falling Error Plots

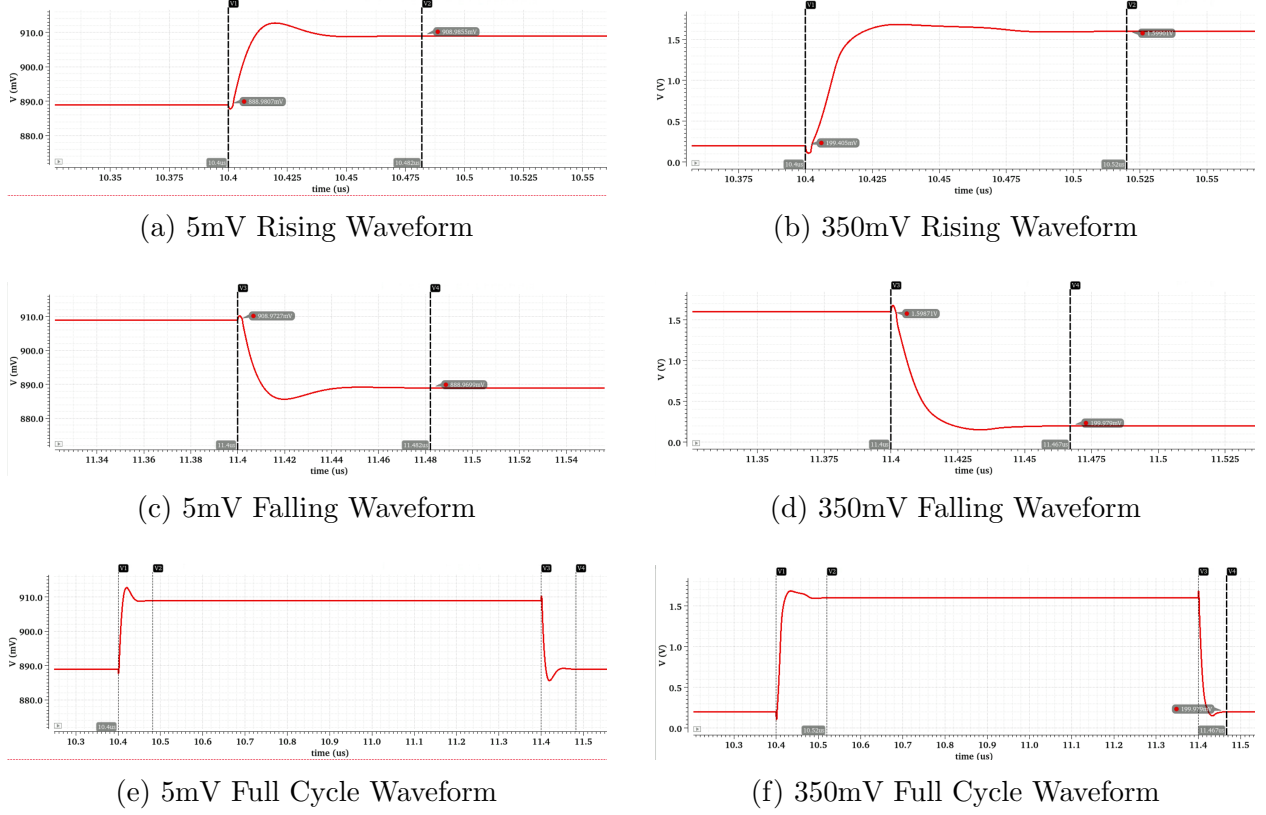


Figure 16: Waveforms at vout

Together the above plots tell similar stories of settling time specifications. Below is a summary of the earlier table condensed to the topics discussed.

Table 4: Summary of Discussion of Specifications

Parameter	Simulated Design	Design Specs (240A)
Settling Time	119ns	$\leq 180ns$
CMRR at DC	77.65dB	$\geq 65dB$
PSRR at DC (VDDL)	57.62dB	$\geq 50dB$
PSRR at DC (VDDH)	53.45dB	$\geq 50dB$
Phase Margin	65.07°	$\geq 45^\circ$
Power Consumption	1.052mW	$\leq 1.5mW$

5 Conclusion

Altogether, the design process was very time consuming, but in the end very worth it and fun at the end of the day.

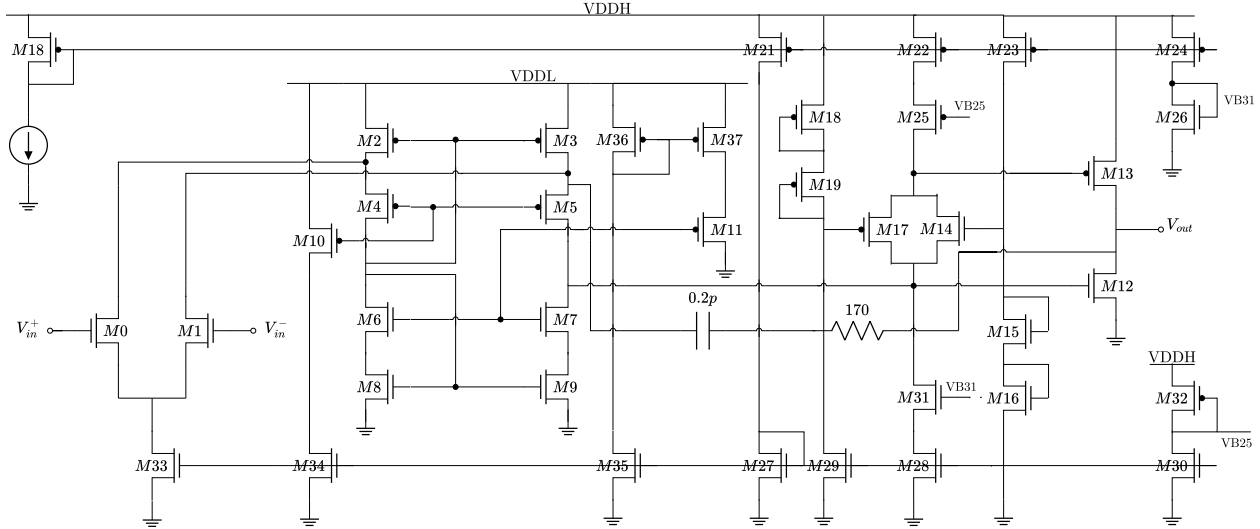


Figure 17: Full schematic without sizings for convenience

My designed amplifier met specifications with reasonably low power draw of $\approx 1mA$. The high gain folded cascode input stage allowed for a high CMRR of $\approx 77dB$ and PSRR for VDDL of $\approx 57dB$, while also helping settling with high gain. The high speed class AB second stage allows for better settling time from slew and reasonable gain. The Ahuja compensation allows for better pole placement and phase margin for better settling time. Finally, the Monticelli biasing and wide-swing cascode biasing allow for high CMRR and PSRR and process and temperature variation tracking.

The process as a whole was incredibly time consuming, but I don't regret putting more and more time into it. About two weeks before the due date, I had a telescopic first stage with a source follower biased class AB second stage and I got it within specifications. I almost decided to call it there, but I thought it would be fun to design a more industry standard and challenging design.

In turn, I really feel like I learned more about integrated circuit design, especially gm/Id methods. To be honest, I didn't fully understand gm/Id from the lab, but the project really taught me how to choose and bias transistors using gm/Id as a heuristic. It probably isn't true, but I truly feel like I ran into every possible problem this project could possibly have. In turn there was a lot of struggle and a lot of learning.

In all, I really enjoyed this project and thought it was worth my time!!!